Name: _____

Investigating density

core practical

Date:

Time:

Total marks available: 31

Total marks achieved: _____

Questions

Q1.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A student determines the density of a liquid.

The student puts an empty measuring cylinder on a balance (Figure 10a).
The student then adds liquid to the measuring cylinder (Figure 10b).

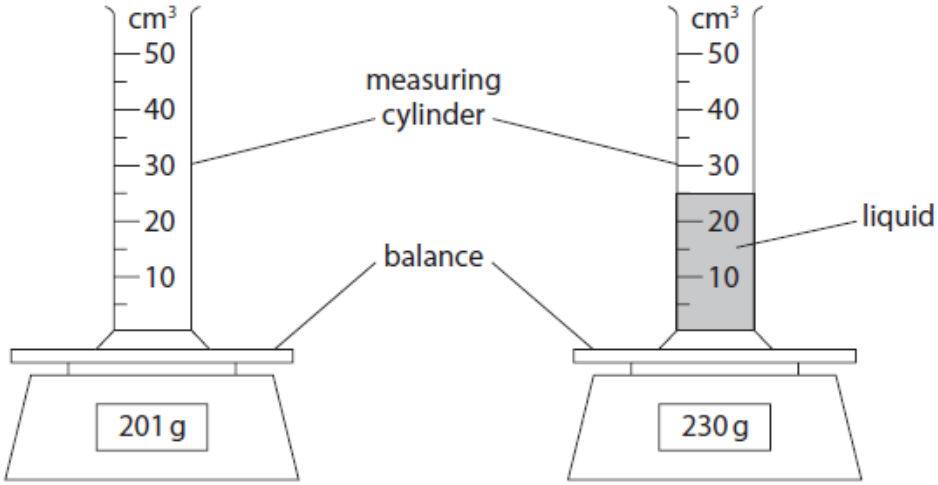

Figure 10a **Figure 10b**

Calculate the mass of liquid added and the volume of liquid added.

Use the information in Figures 10a and 10b.

(i) mass of liquid added = ... g

(1)

(ii) volume of liquid added = ... cm^3

(1)

(iii) Which equation should the student use to calculate the density of the liquid?

(1)

☐ **A** density = mass + volume

☐ **B** density = mass − volume

☐ **C** density = mass × volume

☐ **D** $\text{density} = \frac{\text{mass}}{\text{volume}}$

(iv) State **two** improvements the student could make to this investigation.

(2)

1 .....

.....

2 .....

.....

(Total for question = 5 marks)

Q2.

Figure 19 shows a small piece of copper about 3 cm high.

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Figure 19

A student wants to determine the density of copper.

The student uses a balance to measure the mass of the piece of copper.

(i) Explain how the student could measure the volume of the piece of copper.

(3)

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(ii) The mass of the piece of copper is 0.058 kg.

The volume of the piece of copper is 6.5×10^{-6} m^3.

Calculate the density of copper.

(2)

density of copper = ... kg/m^3

(Total for question = 5 marks)

Q3.

A student is asked to find the density of a metal key.

Figure 8 shows a metal key and some apparatus.

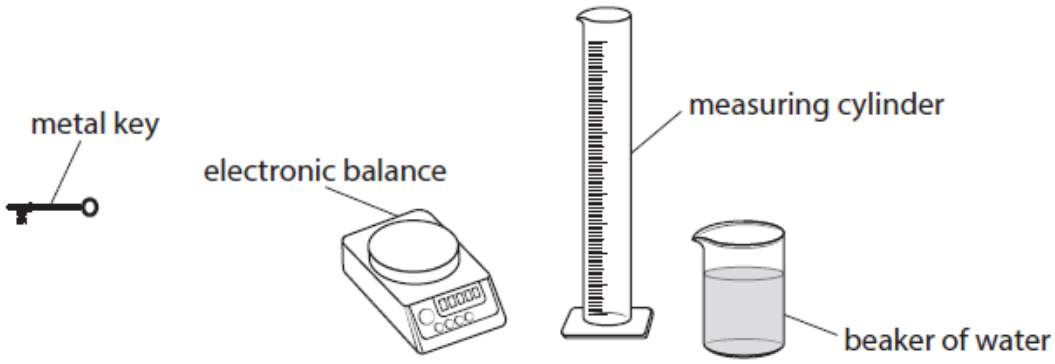

Figure 8

Using the apparatus shown in Figure 8, describe how the student can find the density of the metal key.

$$\text{density} = \frac{\text{mass}}{\text{volume}}$$

(4)

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(Total for question = 4 marks)

Q4.

A piece of wood has a similar shape and volume to the lump of iron.

The density of the wood is 0.82 g / cm^3.

The density of water is 1.00 g / cm^3

Explain why the method used in part (b) cannot be used to determine the mass of the piece of wood.

(2)

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(Total for question = 2 marks)

Q5.

* A student has two metal strips and a ruler, as shown in Figure 22.

A teacher tells the student that

- one metal strip is made of aluminium
- the other metal strip is made of stainless steel.

The student looks up data in a reference book, finding some density values:

density of aluminium = 2710 kg / m^3

density of stainless steel = 7850 kg / m^3

The student has access to more of the same metal strips, if needed, and may ask for any extra measuring devices.

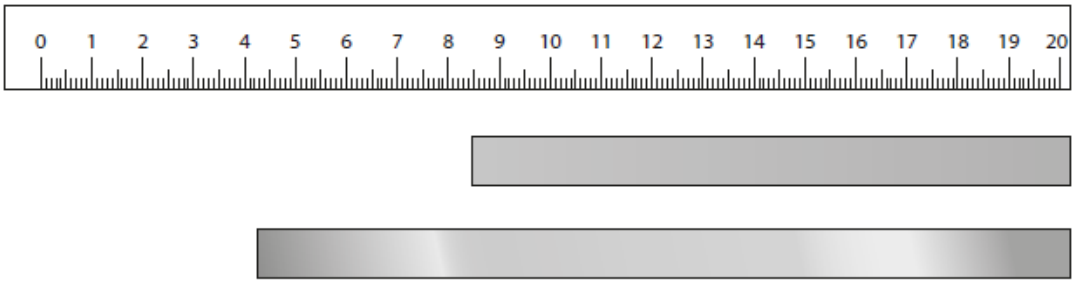

Figure 22

Plan how the student could confirm the teacher's statements, by determining the density of each of the strips as accurately as possible.

(6)

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(Total for question = 6 marks)

Q6.

A student measures the density of glass.

The student has

− a bag of marbles, all made from the same type of glass

− a weighing balance

− a plastic measuring cylinder containing water

Describe how the student could find, as accurately as possible, the density of the glass used for the marbles.

(4)

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(Total for question = 4 marks)

Q7.

A student investigates the density of a copper block and the density of a small stone, as shown in Figure 2.

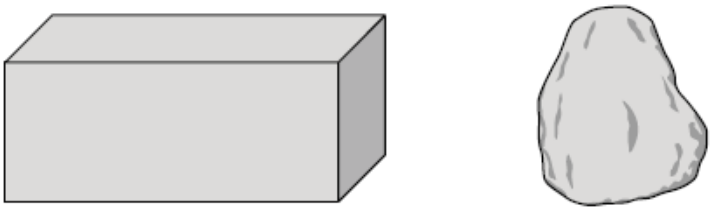

Figure 2

The student found the volume of the copper block by multiplying the area of its base by its height.

The small stone does not have straight sides.

Describe how the student could measure the volume of the small stone. You may use a diagram if it helps your answer.

(3)

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(Total for question = 3 marks)

Q8.

A student determines the volume of a piece of metal by measuring the volume of water that it displaces.

The student wrote the following in his notebook.

> I put some water into a measuring cylinder.
> I put the piece of metal into the water in the measuring cylinder.
> I took the reading of the new water level in the measuring cylinder.
> This was the volume of the piece of metal.

The student's description is incomplete.

Suggest **two** sentences that the student could have included to provide a more complete description of
the correct procedure.

(2)

1 .....

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2 .....

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(Total for question = 2 marks)

Mark Scheme

Q1.

Question Number	Answer	Additional guidance	Mark
(i)	29(g)		**(1)**

Question Number	Answer	Additional guidance	Mark
(ii)	25(cm^3)		**(1)**

Question Number	Answer	Mark
(iii)	**D** density = $\frac{\text{mass}}{\text{volume}}$ **D** is the only correct answer **A** is incorrect because the equation density =mass**+** volume is incorrect **B** is incorrect because the equation density =mass – volume is incorrect **C** is incorrect because the equation density =mass x volume is incorrect	**(1)**

Question Number	Answer	Additional guidance	Mark
(iv)	Any two improvements from:		**(2)**
	use balance that reads to one or more decimal places/more decimal places (1)	Accept use more accurate/precise balance in this context	
	use tare/zero balance for first measurement (1)	Allow reset for tare	
	use measuring cylinder with smaller divisions (1)	Allow more accurate/ different scale / different divisions / thinner measuring cylinder	
	use larger volume of liquid (1)	Allow use more liquid / larger mass of liquid	
	repeat <u>and</u> average (1)		
	read measuring cylinder at eye level (1)	Allow avoid parallax error / read from bottom of meniscus	

Q2.

Question number	Answer	Additional guidance	Mark
(i)	an explanation linking any three of the following : use a measuring cylinder /beaker or use a eureka can /displacement can/container with spout (1) (partly) fill measuring cylinder /beaker (with water) note the reading or fill (eureka) can to spout (1) immerse piece of copper (in water) (1) note difference in readings of water level (in measuring cylinder /beaker) or collect water from spout in a measuring cylinder /beaker (1)	give credit for other acceptable methods If no other marks scored then allow 1 mark for attempt to measure volume directly: e.g. fill copper tube with water, tip out and measure volume or measure dimension(s) of copper tube	(3)

Question number	Answer	Additional guidance	Mark
(ii)	recall and substitution (1) density= $\frac{m}{V}$ (density=) $\frac{0.058}{6.5 (\times 10^{-6})}$ evaluation (1) 8.9×10^{3} (kg/m^{3})	 accept values that round to 8900 e.g. 8923(kg/m^{3}) or 9000 8.9 to any other power of ten gains 1 mark award full marks for correct answer without working.	**(2)**

Q3.

Question Number	Answer	Additional guidance	Mark
	a description including: use the **balance** to find the **mass** of the key (1) (add water and) measure (initial) volume / level of water using the (measuring) cylinder (1) place the key in the measuring cylinder (1) subtract the (two) volumes (to give the volume of the key) (1)	 accept weigh the key using the **balance** allow scales for balance ignore weight accept fill measuring cylinder to a named volume e.g. 100ml accept find the increase/ difference in volume / level of water allow alternative method: use the **balance** to find the **mass** of the key (1) **fill** beaker/eureka can (to the top) with water (1) place key in beaker/eureka can and catch overflow (in measuring cylinder) (1) measure volume / level of water using the (measuring) cylinder (1)	**(4)** **AO1.2**

Q4.

Question number	Answer	Additional guidance	Mark
	an explanation linking density of wood less (than that of water) (1) less (volume of) water displaced (than volume of wood) (1)	 allow wood floats / should be submerged allow wood absorbing water allow (idea of) incorrect volume reading allow (idea that) the volume cannot be measured this way	**(2)** **A02**

Q5.

Question number	Indicative content	Mark
*	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material which is indicated as relevant. Additional content included in the response must be scientific and relevant. **AO1 (strand 2) (6 marks)** Indicative content • measure the length and width of a strip with the ruler / a metre rule • measure the thickness of the strip with a more accurate device e.g. digital callipers OR place 5 (say) of the same strip on top of each other and measure their thickness with the ruler then ÷5 to calculate a single thickness [plus air gap] • measure the mass of a strip with an electronic balance • measure the mass of (say) 5 strips then ÷5 to calculate the mass of one of them • calculate the volume (= l x w x t) in m^3 and the mass in kg • use displacement can/measuring cylinder to find the volume • mass / volume to get density • check if it's near one of the teacher's two values of density given • if it's close / not so far off it's safe to assume that strip is of the identified material • repeat for the other strip • other repeat measurements	**(6)** **AO1**

AO targeting: AO1.2

Level	Mark	Descriptor
	0	• No rewardable material.
Level 1	1-2	• Demonstrates elements of physics understanding, some of which is inaccurate. Understanding of scientific, enquiry, techniques and procedures lacks detail. (AO1) • Presents a description which is not logically ordered and with significant gaps. (AO1)
Level 2	3-4	• Demonstrates physics understanding, which is mostly relevant but may include some inaccuracies. Understanding of scientific ideas, enquiry, techniques and procedures is not fully detailed and/or developed. (AO1) • Presents a description of the procedure that has a structure which is mostly clear, coherent and logical with minor steps missing. (AO1)
Level 3	5-6	• Demonstrates accurate and relevant physics understanding throughout. Understanding of the scientific ideas, enquiry, techniques and procedures is detailed and fully developed. (AO1) • Presents a description that has a well-developed structure which is clear, coherent and logical. (AO1)

Summary for guidance

Level	Mark	Additional Guidance	General additional guidance – the decision within levels e.g. - At each level, as well as content, the scientific coherency of what is stated will help place the answer at the top, or the bottom, of that level.
	0	No rewardable material.	
Level 1	1–2	Additional guidance Partially complete description of a suitable procedure with at least two measurements OR one measurement and another procedural point	Possible candidate responses measure the length measure the width of a strip measure the mass/weight of a strip e.g. repeat measurements
Level 2	3–4	Additional guidance Mostly complete description of a suitable procedure with at least three measurements and some description of processing the results.	Possible candidate responses As above with measure the thickness of the strip calculate the volume (= l x w x t) OR immerse in liquid to get volume
Level 3	5–6	Additional guidance Detailed description of a suitable procedure with all necessary measurements and a clear description of processing the results.	Possible candidate responses As above with extra detail e.g. measure the mass of (say) 5 strips then ÷5 to calculate the mass of one of them detail of obtaining volume by immersion use density =mass /volume check if density value obtained is near one of the teacher's two values

Q6.

Question Number	Answer	Additional guidance	Mark
	A description including: find mass of marble(s) (1) put marble(s) into water (in cylinder) and measure **change** in water level (1) divide mass by volume (1) suitable idea to improve accuracy such as use several marbles (1)	 weigh marble(s) accept volume for water level note level before and after marble(s) added find **volume** of water displaced density = mass/volume in words or symbols subtract mass of bag from total mass of marbles and bag ensure water measured at eye level use appropriately sized measuring cylinder ignore reference to repeating and taking average	**(4)**

Q7.

Question number	Answer	Additional guidance	Mark
	An answer that provides a description by making reference to: • part fill a measuring cylinder with water and record the starting volume (1) • completely immerse the stone in the water and record the final volume of water and stone (1) • volume of stone = final volume – initial volume (1)	accept valid alternative methods, e.g. fill a displacement can until some water overflows/flows out of spout completely immerse the stone in the displacement can and collect the displaced water in a measuring cylinder volume of water displaced = volume of stone	**(3)**

Q8.

Question number	Answer	Additional guidance	Mark
	Any **two** from the following in any order	Answers need not be exactly the same as those given here provided that the meaning is clear.	**(2)** **AO1**
	(I took a) reading of the water level in the measuring cylinder without the metal. (1)	accept measured / read for take a reading accept reading of original level / volume accept starting with a specified amount e.g. 50ml	
	(I made sure that) the metal was fully immersed / submerged (1)	all the metal was under water	

	(I) subtracted the two readings / volumes (1)	took one from the other / found the difference ignore: repeat and/or average other measurements such as mass other methods such as Eureka can ideas of spillage reading from bottom of meniscus	